

Worksheet 14a – Understanding how VAT works

Suppose we measure two variables X_1 and X_2 for each of four items A, B, C and D. The data are given in the following table

Object	Observations	
	x_1	x_2
A	-1	-2
B	-1	1
C	5	3
D	-3	-2

The dissimilarity matrix showing the Manhattan distances between all pairs of objects is given below:

	A	B	C	D
A	0	3	11	2
B	3	0	8	5
C	11	8	0	13
D	2	5	13	0

- (a) Write R code that will produce the exact same dissimilarity matrix as what was shown above.
- (b) By examining the distance (dissimilarity) matrix carefully, determine how many clusters you expect for this small dataset and identify the items within each cluster. It will help you visualize better if you plot the items in a 2D diagram.

- (c) Reorder the items so that similar objects are close to one another. Place the shortest distance close to the top right section and work your way towards the bottom left (for longer distances). The table is drawn below for your convenience:

	Item	Item	Item	Item
Item				
Item				
Item				
Item				

- (d) In the grid in part (c), for every cell with small values (say under 6.5 since range is between 0 to 13), color the cell yellow with a highlighter pen or leave the cell "white". For every cell that has a larger value (say 6.5 or higher), color the cell black. Can you see the number of clusters agree with your answer in part (a)?

- (e) Now use the `fviz_dist(x, show_labels = TRUE)` command in `factoextra` package to see the diagram drawn (VAT approach). Does the diagram match your shadings in part (d)?

R code for parts (a) and (e):

```
A<-c(-1,-2)
B<-c(-1,1)
C<-c(5, 3)
D<-c(-3,-2)
mymat<-rbind(A,B,C,D)
getdist<-dist(mymat, method="manhattan")
distmat<-as.matrix(getdist)
fviz_dist(getdist, show_labels=TRUE)
```